Physics 4B Spring 2026
Week 9 Discussion

TA: Jiashu Han

May 27, 2026

1. **Brainwarmer**[1]. What is the increase in entropy if one gram of ice at 0°C is melted and heated to 50°C?

   Solution: We can compute the entropy increases of the two processes separately and then add them. The entropy increase during the melting (at constant temperature) is

   $$\Delta S_{\text{melt}} = \frac{\delta Q_{\text{melt}}}{T_{\text{melt}}} = \frac{mL}{T_{\text{melt}}} = \frac{(0.001\text{ kg})(3.34\times 10^5\text{ J/kg})}{273\text{ K}} = 1.22\text{ J/K} \tag{1}$$

   The entropy increase during the heating is

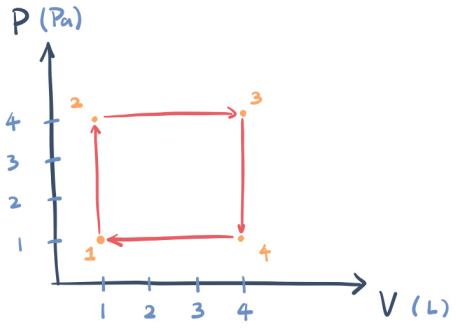
   $$\begin{aligned}\Delta S_{\text{heat}} &= \int_{T_i}^{T_f}\frac{dQ}{T} = \int_{T_i}^{T_f}\frac{mc\,dT}{T} = mc\ln\left(\frac{T_f}{T_i}\right)\\ &= (0.001\text{ kg})(4184\text{ J/(kg}\cdot\text{K)})\ln\left(\frac{323\text{ K}}{273\text{ K}}\right) = 0.7\text{ J/K}\end{aligned} \tag{2}$$

   The total change in entropy is therefore

   $$\Delta S = \Delta S_{\text{melt}} + \Delta S_{\text{heat}} = 1.92\text{ J/K} \tag{3}$$

2. **Heat engine efficiency**[2]. A heat engine goes through the rectangular loop $1\to 2\to 3\to 4\to 1$ with corners 1: (1 Pa; 1 L); 2: (4 Pa; 1 L); 3: (4 Pa; 4 L); 4: (1 Pa; 4 L). In addition, $T_1 = 100$K and $\gamma = c_p/c_v = 1.4$ where c_p and c_v are specific heats per mole. What is the efficiency of the engine? *Hints: Assume only that $c_p = c_v + R$, deduce c_v in terms of nR, and use the ideal gas law repeatedly.*

   Solution: The PV diagram of this heat engine looks like:



[1] Shankar 11.29
[2] Shankar 11.20

To find the efficiency, we need to figure out the total work it outputs in one cycle, and the total heat it absorbs over the cycle. The 1-2 and 3-4 segments are isochoric processes, with no work done, while the 2-3 and 4-1 segments are isobaric processes, so the total work done is

$$\begin{aligned} W &= W_{23} + W_{41} = P_2(V_3 - V_1) + P_4(V_1 - V_4) \\ &= (4\text{ Pa})(3 \times 10^{-3}\text{ m}^3) + (1\text{ Pa})(-3 \times 10^{-3}\text{ m}^3) = 0.009\text{ J} \end{aligned} \tag{4}$$

For the total heat absorbed, we need to figure out the heat capacities first. Since $\gamma = c_p/c_v = 1 + \frac{R}{c_v} = 1.4$, the molar heat capacities are

$$c_v = \frac{5}{2}R,\ c_p = \frac{7}{2}R \tag{5}$$

We can then calculate the heat absorbed for each segment

$$Q_{12} = nc_v(T_2 - T_1) = \frac{5}{2}nR(T_2 - T_1) = \frac{5}{2}(P_2V_2 - P_1V_1) = 0.0075\text{ J} \tag{6}$$

$$Q_{23} = nc_p(T_3 - T_2) = \frac{7}{2}nR(T_3 - T_2) = \frac{7}{2}(P_3V_3 - P_2V_2) = 0.042\text{ J} \tag{7}$$

$$Q_{34} = nc_v(T_4 - T_3) = \frac{5}{2}nR(T_4 - T_3) = \frac{5}{2}(P_4V_4 - P_3V_3) = -0.03\text{ J} \tag{8}$$

$$Q_{41} = nc_p(T_1 - T_4) = \frac{7}{2}nR(T_1 - T_4) = \frac{7}{2}(P_1V_1 - P_4V_4) = -0.0105\text{ J} \tag{9}$$

The efficiency is then the work divided by the heat absorbed into the system:

$$\eta = \frac{W}{Q_{\text{in}}} = \frac{W}{Q_{12} + Q_{23}} = 18\% \tag{10}$$

3. **Free expansion and adiabatic expansion**[3]. One mole of an ideal monoatomic gas, thermally insulated, is restricted by a partition to a volume V_0 of a cylinder whose total volume is $3V_0$. Its pressure is P_0.

   (i) When the partition is removed and the gas fills the whole box, what is the change in entropy of the gas and of the universe?

   (ii) What is the ratio of microscopic states available after the partition is removed to before?

   (iii) Suppose the partition is replaced by a movable piston which slowly lets the gas expand and provides an equal and opposite pressure until the volume becomes $3V_0$. (The system is still in complete thermal isolation.) What is the change in entropy now?

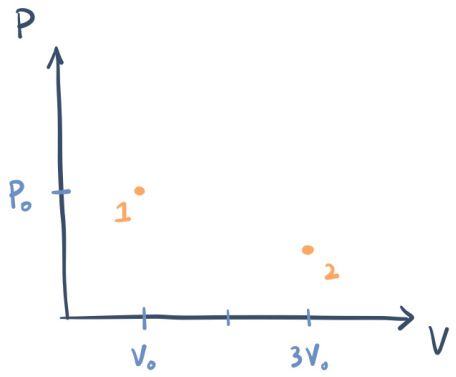
[3]Shankar 11.31

(iv) What is the work done by the gas in terms of V_0 and the initial pressure P_0?

(v) What is the change in internal energy?

Solution:

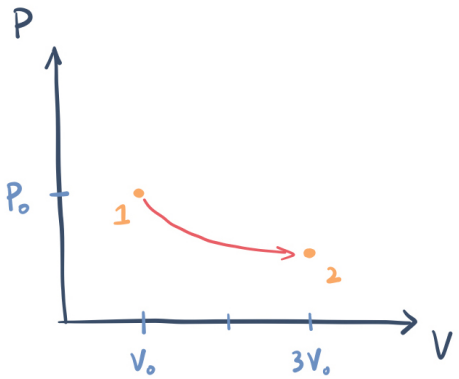
(i) The PV diagram for the free expansion is



How do we know the final state has to be where it is? One important observation is that there's no change in the internal energy of the system ($\Delta U = 0$) - it's thermally isolated ($\delta Q = 0$) and no work was done by the gas ($\delta W = 0$) because the gas did not push on anything and changing its volume during the expansion. Therefore, the temperature of the gas remains the same, and $(P_1, V_1) = (P_0, V_0)$ and $(P_2, V_2) = (P_0/3, 3V_0)$ should be on the same isotherm.

To find the change in entropy of the gas, we imagine there's a path from 1 to 2 (the simplest of which is isothermal), and integrate $dS = dQ/T$ along that path. We can do this because entropy is a state variable, and ΔS does not depend on the path taken between the two points.

We are going to do an isothermal process, as shown below



for which we have a nonzero δQ, which is equal to the work done by the gas due to $\Delta U = 0$, so we can replace dQ with PdV:

$$\Delta S_{\text{gas}} = \int_1^2 \frac{dQ}{T} = \int_1^2 \frac{P}{T} dV = nR \int_{V_0}^{3V_0} \frac{dV}{V} = nR \ln\left(\frac{3V_0}{V_0}\right) = R \ln 3 \quad (11)$$

where we plugged in $n = 1$. There's no change to the part of the universe outside the gas, so the entropy change of the universe is the same:

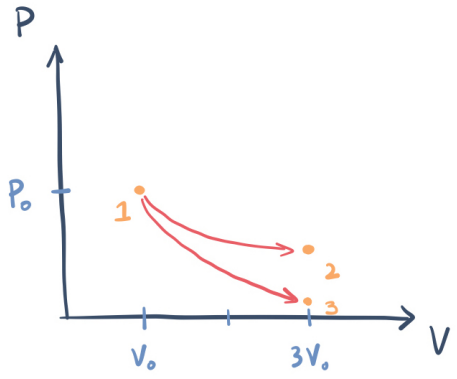
$$\Delta S_{\text{universe}} = R \ln 3 \quad (12)$$

(ii) Since $S = k \ln \Omega$, the ratio of microscopic states available is

$$\frac{\Omega_f}{\Omega_i} = \frac{e^{S_f/k}}{e^{S_i/k}} = e^{\Delta S/k} = e^{N_A k (\ln 3)/k} = 3^{N_A} \quad (13)$$

where N_A is Avogadro's number.

(iii) This question is conceptually tricky. By making the process slow and reversible, it is a well defined curve on the PV diagram, but it will not be the same curve as the isotherm we chose to use in (i), because now we cannot simply conclude there's no change in the internal energy. In fact, the gas does work to push the piston out ($\delta W \neq 0$), while the system is kept thermally isolated (i.e. $\delta Q = 0$), so there's definitely a change in temperature and internal energy, and we should end up with a pressure different from $P_0/3$. This new process is shown as 1→3 in the diagram below



The 1→3 curve brings the gas to a lower temperature and pressure compared to the free expansion scenario in (i).

Given that the process is adiabatic and reversible, we can apply the $\int \frac{dQ}{T}$ formula directly. Since $\delta Q = 0$, the entropy change is simply

$$\Delta S = \int_1^3 \frac{dQ}{T} = 0 \quad (14)$$

for both the gas and the universe. This may feel somewhat unintuitive, but adiabatic curves on the PV diagram are curves of constant entropy.

(iv) We can certainly use the adiabatic relation $PV^\gamma = \text{constant}$ and $\delta W = PdV$ to figure out the work done by the gas, but there's a much easier way.

Since there's no heat flow, by the first law, the work done by the gas is equal to the (negative) change in the internal energy

$$\begin{aligned} W &= -\Delta U = -\frac{3}{2}nR(T_2 - T_1) = \frac{3}{2}nRT_1\left(1 - \frac{T_2}{T_1}\right) \\ &= \frac{3}{2}P_0V_0\left(1 - \frac{T_2}{T_1}\right) \end{aligned} \tag{15}$$

We don't know the temperature T_2, but we can use the adiabatic relation

$$P_1V_1^\gamma = P_2V_2^\gamma \tag{16}$$

to replace the temperature ratio with something we do know. This can be rewritten as

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} \tag{17}$$

by plugging the ideal gas law. The work is then

$$\begin{aligned} W &= \frac{3}{2}P_0V_0\left(1 - \frac{T_2}{T_1}\right) = \frac{3}{2}P_0V_0\left(1 - \left(\frac{V_1}{V_2}\right)^{\gamma-1}\right) \\ &= \frac{3}{2}P_0V_0\left(1 - \left(\frac{V_0}{3V_0}\right)^{2/3}\right) = \frac{3}{2}P_0V_0\left(1 - 3^{-2/3}\right) \end{aligned} \tag{18}$$

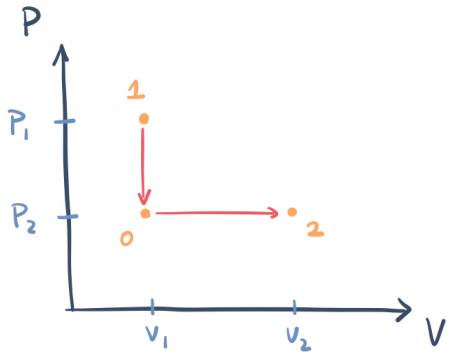
where we used $\gamma = \frac{c_p}{c_v} = \frac{5}{3}$ for a mono-atomic gas.

(v) As mentioned in the previous part, $\Delta U = -W = \frac{3}{2}P_0V_0\left(3^{-2/3} - 1\right)$.

4. **Entropy of ideal gas**[4]. Consider a mole of a gas initially at $1 \equiv (P_1, V_1)$ and finally at $2 \equiv (P_2 < P_1, V_2 > V_1)$. Since $S_2 - S_1$ is path independent, choose the simple path by first lowering pressure at constant volume and then increasing volume at constant pressure. Let $0 \equiv (P_0, T_0)$ be the intermediate point you go through (Thus $P_0 = P_2$ and $V_0 = V_1$). Show that $S_2 - S_1 = c_p \ln\left(\frac{T_2}{T_1}\right) - R\ln\left(\frac{P_2}{P_1}\right)$. Show that if 1 and 2 lie on an adiabatic curve, this difference vanishes. Assume $c_p = c_v + R$ but not a particular value to c_v.

Solution: The PV diagram looks like

[4]Shankar 11.34



For the 10 segment, the change in entropy is

$$\Delta S_{10} = \int_1^0 \frac{dQ}{T} = nc_v \int_{T_1}^{T_0} \frac{dT}{T} = nc_v \ln\left(\frac{T_0}{T_1}\right) \tag{19}$$

Similarly, the entropy change for the 02 segment is

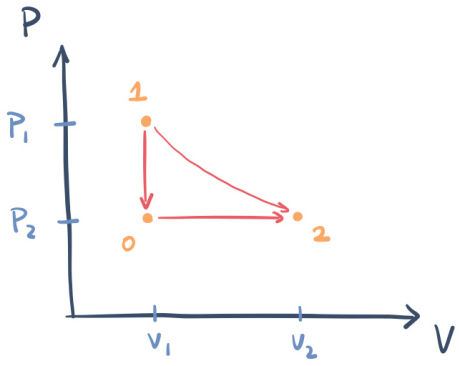
$$\Delta S_{02} = \int_0^2 \frac{dQ}{T} = nc_p \int_{T_0}^{T_2} \frac{dT}{T} = nc_p \ln\left(\frac{T_2}{T_0}\right) \tag{20}$$

The total entropy change is then

$$\begin{aligned}
\Delta S = S_2 - S_1 = \Delta S_{10} + \Delta S_{02} &= nc_v \ln\left(\frac{T_0}{T_1}\right) + nc_p \ln\left(\frac{T_2}{T_0}\right) \\
&= nc_p \ln\left(\frac{T_0}{T_1}\right) - nR \ln\left(\frac{T_0}{T_1}\right) + nc_p \ln\left(\frac{T_2}{T_0}\right) \\
&= nc_p \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2 V_1}{P_1 V_1}\right) \\
&= nc_p \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2}{P_1}\right)
\end{aligned} \tag{21}$$

where $n = 1$.

If we connect 1 and 2 with an adiabatic curve



then we have

$$P_1 V_1^\gamma = P_2 V_2^\gamma \tag{22}$$

which we can use the ideal gas law to rewrite as

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma \tag{23}$$

Plug this into our result for the previous part, we get

$$\begin{aligned} & c_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) \\ = \; & c_p \ln\left(\frac{P_1}{P_2}\right)^{(1-\gamma)/\gamma} - R \ln\left(\frac{P_2}{P_1}\right) \\ = \; & c_p \frac{1-\gamma}{\gamma} \ln\left(\frac{P_1}{P_2}\right) - R \ln\left(\frac{P_2}{P_1}\right) \\ = \; & 0 \end{aligned} \tag{24}$$

where in the last line we used $\gamma = c_p/c_v$ and $c_p = c_v + R$.